

**DEPARTMENT OF CHEMICAL ENGINEERING
NORTH CAROLINA STATE UNIVERSITY**

CHE 331 QUIZ #2

This is a 30 minute closed-book quiz. Write your name and answer the quiz problems related to your experiment. Turn in this question sheet with your answer.

Name: _____

Evaluation of Fluidized Bed Characteristics Experiment

Problem 1 (30%) Examine the plot shown in Figure 1 describing pressure loss across a fluidized bed of an experimental catalyst material as a function of superficial gas velocity (u_0). **(a, 10%)** What would you estimate is the minimum fluidization velocity for the experimental particle? **(b, 20%)** Why is the pressure drop behavior different depending on whether data is taken while increasing fluid speed or taken while decreasing fluid speed?

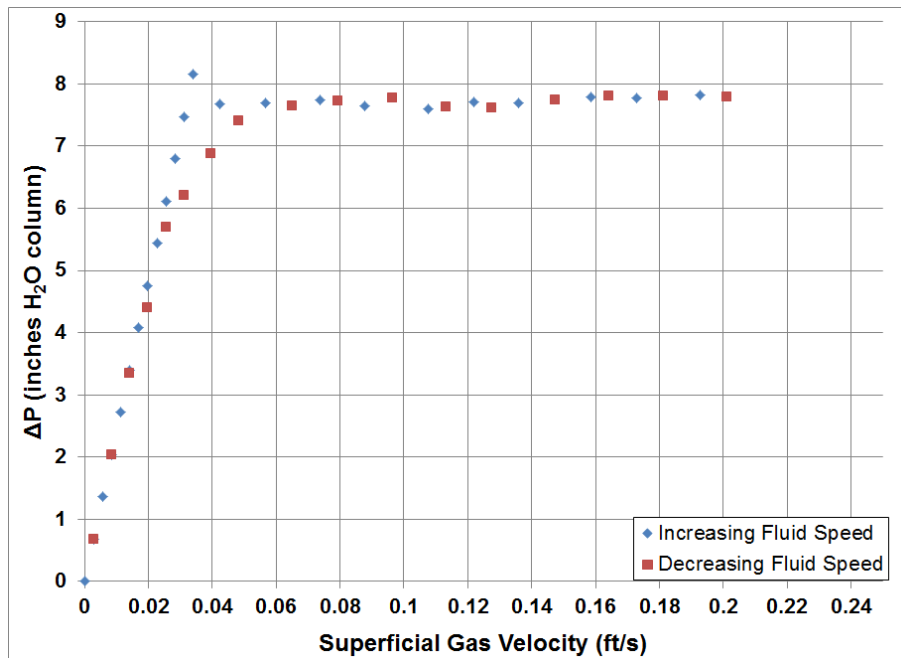


Figure 1. Fluidization curve for experimental catalyst particle.

Problem 2 (20%) In your own words, describe the force balance on a particle in a fluidized bed.

Problem 3 (20%) The Geldart classification of particles is based on two properties of the particle. What are these properties?

Problem 4 (30%) (a, 25%) Sketch a clearly-labeled plot describing the relationship between heat transfer coefficient (α_{exp}) for the fluidized bed and superficial gas velocity. **(b, 5%)** In which fluidization regime did you find the highest heat transfer coefficient? Why do you think this is the case?